

Instruction for the use of Perl scripts

Basic setup

- all scripts depend on subroutines stored in the perl module Biomodule.pm.
- place the Biomodule.pm and all WICKERsoft scripts into a dedicated directory which is in a common path (e.g. /usr/local/bin/) so that you can run them from anywhere in your system.
- make sure that all scripts are executable. if necessary, make them executable with `chmod 755 <script_name>`
- change the path in the beginning of all WICKERsoft scripts to that directory

example, how the header should look:

```
#!/usr/bin/perl
```

```
use lib '/usr/local/bin/';
```

```
use Biomodule;
```

Additional software needed:

From standard Ubuntu repositories:

Emboss

Clustalw

blastall / ncbi-blast+-legacy (local stand-alone blast program)

perl-tk

dotter

libgd-graph-perl

libgd-perl

libgd-text-perl

Blast programs

the Perl scripts described here use the **blastall** program from ubuntu repositories. Searchable databases were produced with the format **formatdb** that come along with the package. By default they are placed in a directory /home/wicker/data/databases/. This path has to be adjusted in the scripts.

Blast requires the BLOSUM62 matrix which is also provided in the repository.

All original Perl scripts used here show a **usage manual** when the program name is typed without arguments (this shows a list of the required arguments)

Procedures for this study are listed below

If there are questions or problems, contact wicker@botinst.uzh.ch

Chromosome colienarity analysis

This protocol is used to identify colinear chromosomal region, TE presence/absence polymorphisms (PAPs) and deletions

In this study this procedure for the chromosome conservation plots and identifications of deletions and TE insertions in Supplementary Figure S18.

Alignments of megabase-sized chromosomal fragments

chromosomes are aligned with the Perl script **blast_compare_chromosome**. This runs blastn searches of a given chromosome from species/haplotype against the homologous chromosome of the other species/haplotype. It splits the sequence 1 in user-defined (e.g. 1 kb) windows with a user-defined sliding step between windows and uses them in blastn searches against the other chromosome.

The blast output is parsed and the position of the best hit on sequence 2 is recorded. Furthermore, the top alignment is extracted and the numbers of SNPs is recorded. Transitions and transversions are treated separately in case they are used for molecular dating (this was not used in the study).

The program produces a log file with the name prefix "log_chr_comp_"

The header line has names and sizes of chromosomes (or chromosome segments) used

The columns in the log file are as follows:

1. ID of segment compared
2. start position of segment used for blastn on Bgt_96224_v316_chr11
3. start position of top hit on Bgs_1451_chr11
4. number of aligned bases
5. Sequence identity
6. Number of transitions ($A \leftrightarrow G$ and $C \leftrightarrow T$)
7. Number of transversions (all other substitutions)
8. Divergence time estimate based on a substitution rate of $1.3E-8$
9. Standard deviation of divergence time estimate

Chromosome dot plots

The log files produced by **blast_compare_chromosome** are visualized as dot plots with the Perl script **visual_chromosome_compare_dotplot**. This uses the Perl-Tk module for an interactive GUI. It produces a simple dot plot matrix from the positions of the best blastn hits. For the plots in the figure, the option "color_hits" and color_contrast can be used for the adjustment of the color scale for sequence identity.

Procedures used in this study are described in more detail below

Procedure used in this study

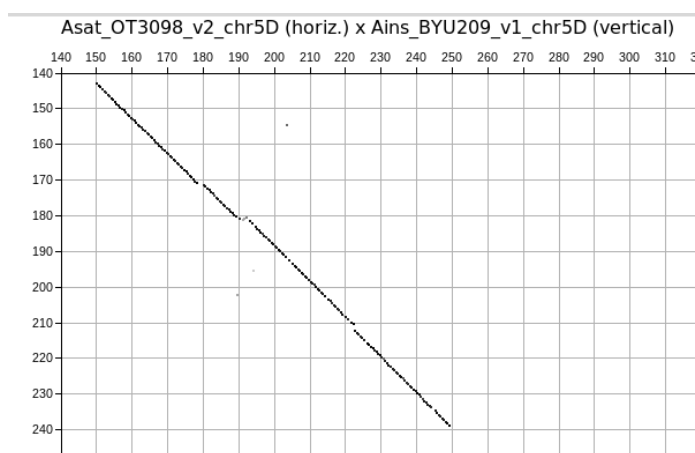
Comparison of centromeric and peri-centromeric regions

Step 1: Identification of large colinear segments

Large colinear segments are identified in a first overall comparison in which the centromeric regions (centromeres plus ~50 Mb of flanking regions) were compared with **blast_compare_chromosome**. Log files were visualized with **visual_chromosome_compare_dotplot**.

Identification of TE presence/absence polymorphisms (PAPs) was done on chromosomal segments of 5-30 Mb which show good colinearity so that they can be aligned reliably. Such colinear fragments are identified in a first overall comparison in which the centromeric regions (centromeres plus ~50 Mb of flanking regions) were aligned with the script **blast_compare_chromosome** using a segment size of 1000 and a sliding step of 500 kb. Based on this, target segments of 5-30 Mb that show no inversions of large non-colinear segments were selected

An example plot shows comparison of chromosome 5 of *Avena sativa* and *A. insularis* for the region 150-250 Mb



Step 2: Based on the first global alignment colinear segments (in this study including the centromere were defined. Selected colinear segments are excised from the chromosomes of both species with the script **assemble**. Positions (in Mb) of the selected segments are provided in **positions_colin_segments**.

Step 3: The excised colinear fragments from the two species were aligned again with **blast_compare_chromosome** using a segment size of 1000 and a sliding step of 1000 bp (in order to generate quasi gap-free alignments subsequently). This was done in both directions, as the following steps only identifies insertions in one direction (e.g. insertions in species 2, when species 1 is aligned with species 2).

Step 4: the program **TE_PAP_from_log_chr_comp** was used to identify large (i.e. TE-sized) insertions in species 2. The program first scans the log_file of blast_compare_chromosome for colinear regions using the following criteria:

1. All blast hits in reverse orientation are dismissed
2. Non-colinear hits are identified by calculating the median over 30 position of hits in species 2. All hits that deviate from the median by more than 35 kb are dismissed as hits to non-colinear loci.
3. the remaining filtered blast hits are then screened for insertion in species 2 that are between 3kb and 17 kb.c
4. The program runs a dotplot alignments using the software dotter (unutu repositories) for visual inspection of the candidate InDel regions and asks the used whether the InDel candidate should be kept
5. If the region is deemed a true indel, the program then runs a local alignment using the Smith-Waterman algorithm implemented in the program water (EMBOSS package) and excises the largest insertion in species 2.

Step 5. The isolated insertion sequences were used as queries against the nrTREP25 TE database to identify homologies with know TE families

Step6. The program **png_blast_coverage_TE_all** is used to manually examine the blast outputs and distinguish insertions of full-length TE from deletions of random segments.

- The program produces a graphical representaion of blast hits in linear and dot plot format

This allows easy visual inspection if an InDel corresponds to an insertion of a full-length element (i.e. the full TE consensus aligns with the InDel).

- If an InDel is not classified as insertion in sequence 2, it is classified as deletion in sequence 1.

(Since the regions examined are comprised almost exclusively of retrotransposons, a screening for genes was not done)

- In case of deletions, the program calculates the portion of different TE families in each query sequence based on the blastn results

- the program produces a log file (**log_TE_fam_cvg_insertions_INS**)

Step 7. Final annotation files are produced with the script **log_TE_fam_cvg_to_anno**

This script produces annotations for Insertions and deletions on the selected segments (provided in **annotation_InDels_selected_segments.tar.gz**) as well as their absolute positions on chromosomes (provided in **annotation_InDels_abs_chr_pos.tar.gz**).

Annotations were then visualized with **visual_chromosome_compare_segment**. A screenshot of the display was used for Fig S18b. The table **log_TE_fam_cvg_insertions_INS_eval_InDel_counts** contains summary of numbers and lengths of characterized InDels used for Fig S18c and S18d.

Step 8. The perl script **TE_PAP_eval_TE_cvg** used the log file **log_TE_fam_cvg_insertions_INS** to calculate contributions of all identified TE families to all segments analyzed. These were then used to quantify contributions of Centromere specific TE families to deleted segments (table **log_TE_fam_cvg_insertions_INS_eval**, Fig S18e).